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Course Code: CSA0801

Slot: C

Course Name: PYTHON PROGRAMMING

Course Faculty: Dr.U.SAMSON EBENEZAR

PROJECT TITLE: HANDWRITTEN DIGIT RECOGNITION USING DEEP LEARNING

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

MODULE PHOTO

**MODULE 2
CNN MODEL DEVELOPMENT**

- Build a CNN model for handwritten digit recognition.
- Add convolution and pooling layers to extract features.
- Use ReLU activation for better learning.
- Add a Flatten and Dense layer for classification.
- Use Softmax to predict digits from 0-9.

**MODULE 2
CNN Model Development**

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Conv2D, MaxPooling2D, Flatten, Dense, Softmax

# Load and preprocess data
(x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train = x_train.reshape([x_train.shape[0], x_train.shape[1], x_train.shape[2], 1])
x_test = x_test.reshape([x_test.shape[0], x_test.shape[1], x_test.shape[2], 1])
x_train = x_train.astype('float32') / 255
x_test = x_test.astype('float32') / 255

# Build the CNN model
model = Sequential()
model.add(Conv2D(32, kernel_size=(3, 3), activation='relu', input_shape=(28, 28, 1)))
model.add(MaxPooling2D(pool_size=(2, 2), activation='relu'))
model.add(Flatten())
model.add(Dense(100, activation='relu'))
model.add(Dense(10, activation='softmax'))

# Compile the model
model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
```

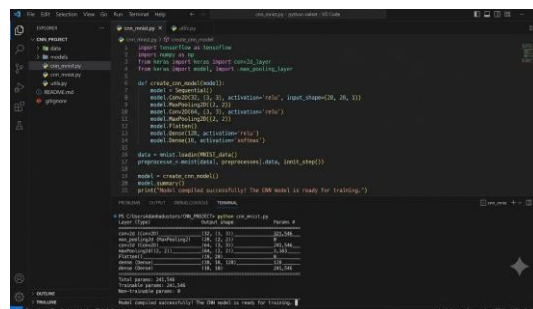
Model: 'sequential'

Layer (type)	Output Shape	Param #
Conv2D (Conv2D)	(None, 32, 32, 1)	32
MaxPooling2D (MaxPooling2D)	(None, 16, 16, 1)	0
Flatten (Flatten)	(None, 256)	0
Dense (Dense)	(None, 100)	25,700
Dense (Dense)	(None, 10)	1,010
Total		26,732

Model compiled successfully!
The CNN model is ready for training.

MODULE WORK IN THE PROJECT

INDIVIDUAL CONTRIBUTION



PRESENTATION IMAGE

REVIEW-1



REVIEW-2



Project Description: Module 2:CNN MODEL DEVELOPMENT

In this project, I was responsible for the design and development of the Convolutional Neural Network (CNN) model for handwritten digit recognition. I developed the CNN architecture by selecting suitable convolutional, pooling, and fully connected layers to effectively extract important features from digit images. I implemented the model using deep learning techniques and configured the required activation functions, loss function, and optimizer. I also prepared the model for training and validation to ensure accurate digit classification. The developed CNN model forms the core of the system, enabling it to recognize handwritten digits from input images

Student Signature

Guide Signature